



ELEVVO INTERNSHIP 2025 (TASK-04)

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Time Series Breakdown of Walmart Retail Sales

1. Objective

The goal of this task was to analyze Walmart's historical sales data, identify overall trends, seasonal behavior, and fluctuations, and to build a simple baseline forecast. The analysis helps in understanding sales patterns that can guide decisions such as promotions, inventory planning, and resource allocation.

2. Dataset Overview

- Source: Walmart Sales Forecasting dataset (Kaggle).
- Size: 421,570 rows × 5 columns.
- Columns: Store, Dept, Date, Weekly_Sales, IsHoliday.
- Date Range: February 2010 to October 2012.
- Frequency: Weekly data, aggregated into monthly totals for time series analysis.

3. Steps Performed

Data Preparation

- Loaded the dataset and converted the Date column into datetime format.
- Sorted the dataset by Date.
- Aggregated weekly sales into monthly totals for clearer trend analysis.

Exploratory Analysis and Visualization

1. Overall Sales Trend

- Plotted the monthly total sales.
- Insight: Sales showed fluctuations with noticeable peaks, suggesting seasonal influences such as holidays or promotions.

2. Moving Averages

- Computed and plotted 3-month and 6-month moving averages.
- Insight: Moving averages smoothed out short-term spikes, highlighting the longer-term upward and downward shifts.

3. Seasonality Analysis

- Decomposed the time series into trend, seasonality, and residuals.
- Insight: Strong seasonal patterns were observed, with sales consistently rising around specific months, likely holiday-driven.

4. Category and Regional Breakdown (if applicable)

- Sales were grouped by department or store to check variation across different segments.
- Insight: Some regions and product categories experienced sharper seasonal peaks than others.

Forecasting

- Implemented simple forecasting techniques:
 - Rolling mean forecast.
 - Exponential Smoothing using Holt-Winters method.
- Forecasted sales for the next 3–6 months.
- Evaluated forecast accuracy using MAE and RMSE.
- Insight: Simple exponential smoothing performed better than rolling averages but still struggled to fully capture large holiday spikes.

4. Key Findings

- Sales patterns are seasonal, with strong peaks around holidays.
- Moving averages highlight general upward and downward sales trends.
- Forecasting methods provide a reasonable baseline but underpredict holiday spikes.

5. Recommendations

1. For more accurate forecasting, advanced models such as SARIMA, Prophet, or LSTM should be applied.
2. Holiday and promotion variables should be explicitly included in forecasting models.
3. Inventory and staffing should be increased in months with consistently higher seasonal demand.
4. Further segmentation by store and department can help identify which regions or product categories drive seasonal peaks.

6. Deliverables

- Jupyter Notebook (`D.A Elevvo Task 4.ipynb`).
- Cleaned dataset (`monthly_total_sales.csv`).
- Plots:
 - Overall sales trend.
 - Moving averages.
 - Seasonal decomposition.
 - Forecast plots.
- Report (`Task 4 Elevvo Report.pdf`).
- Requirements file (`requirements.txt`).

